School of Future Tech

Case Study Report

on

Courier & Parcel Tracking System

by

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1. Introduction to the Case Study

Courier services have become an essential part of modern logistics. People frequently send parcels, documents, and products from one place to another, and it is important for both the courier company and the customer to know the status of the parcel during the delivery process.

In many small courier operations, parcel information is still recorded manually, which can lead to confusion, delays, or loss of information. Because of this, a simple digital system can help in maintaining parcel records and tracking their delivery progress.

This case study focuses on the development of a Courier & Parcel Tracking System using C++. The system allows courier staff to register parcels, update their delivery status, confirm deliveries, and view all parcel records. Customers can also track their parcel using a unique tracking ID.

The purpose of this case study is to demonstrate how basic programming concepts such as classes, vectors, loops, and menu-driven programs can be used to create a practical application that solves a real-world problem.

2. Problem Statement / Case Background (Abstract)

Background

Courier companies deal with many parcels every day. Without a proper tracking system, it becomes difficult to maintain records of parcels, delivery addresses, and their current delivery status. Customers also want to know where their parcel currently is and when it will be delivered.

A simple digital system can help manage this information efficiently and reduce manual work.

Abstract

This case study presents the design and implementation of a **Courier & Parcel Tracking System developed in C++**. The system enables courier staff to register parcels by entering sender information, receiver information, dispatch address, and delivery address. Each parcel is assigned a unique tracking ID that can be used to monitor its status.

The system allows staff to update parcel status such as **In Transit**, **Out for Delivery**, and **Delivered**. Customers can enter the tracking ID to view the current status and location of their parcel. The program uses object-oriented programming concepts and dynamic data storage to maintain parcel records efficiently.

3. Case Study Design

The Courier & Parcel Tracking System is designed as a menu-driven console application where users can select different options to perform operations.

The system has two main user roles:

1. Courier Staff

Courier staff members are responsible for managing parcel information. The system allows them to:

- Register a new parcel
- Update the parcel tracking status
- Confirm parcel delivery
- View all registered parcels

During parcel registration, the staff member enters the sender name, receiver name, dispatch address, and delivery address. The system then generates a unique tracking ID for the parcel.

2. Customer

Customers can track the status of their parcel using the tracking ID. The system displays information such as:

- **Sender name**
- **Receiver name**
- **Dispatch address**
- **Delivery address**
- **Current parcel location**
- **Delivery status**

System Workflow

1. The program starts with a main menu.
2. The user chooses between Courier Staff or Customer.
3. Staff members can register parcels and update delivery status.
4. Customers can check parcel details using a tracking ID.
5. The system continues running until the user selects the exit option.

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

The implementation of this system uses several programming techniques and concepts.

1. Object-Oriented Programming

The system uses a Parcel class to represent parcel information. This class stores details such as sender name, receiver name, dispatch address, delivery address, and status. Using a class makes the program more organized and easier to manage.

2. Dynamic Data Storage

The program uses the vector data structure in C++ to store parcel records dynamically. This allows the system to store multiple parcels without knowing the exact number in advance.

3. Searching Algorithm

A simple linear search algorithm is used to find parcels by their tracking ID. The program checks each stored parcel until it finds the matching ID.

4. Menu-Driven Program Structure

The system uses loops and switch statements to create an interactive menu where users can choose different options. This structure makes the program easy to navigate and user friendly.

Technology Stack Used

Programming Language: C++

Development Environment:
Visual Studio Code / CodeBlocks / any C++ compiler

Libraries Used:

- iostream for input and output
- vector for dynamic storage
- iomanip for formatted output

5. Problem Statement / Case Study Implementation Details and Snapshots.

The system is implemented as a console-based C++ program with multiple functions that handle different tasks.

Main Components of the System

1. **Parcel Class**

Stores all parcel information such as ID, sender name, receiver name, addresses, location, and delivery status.

2. **Parcel Registration Module**

Allows staff to register a new parcel by entering sender details, receiver details, dispatch address, and delivery address. A unique tracking ID is generated automatically.

3. **Status Update Module**

Allows courier staff to update the parcel status during the delivery process. For example:

- In Transit
- Out for Delivery

The current location of the parcel can also be updated.

4. **Delivery Confirmation Module**

Once the parcel reaches the destination, the staff can confirm the delivery and the system updates the parcel status to Delivered.

5. **Parcel Tracking Module**

Customers can enter the tracking ID to view the current parcel details and status.

Example Output

Parcel Registration Example:

Enter Sender Name: Aman Sharma

Enter Receiver Name: Rahul Verma

Enter Dispatch Address: Delhi Sorting Center

Enter Delivery Address: MG Road Pune

Parcel Registered Successfully

Tracking ID: 1001

Customer Tracking Example:

Enter Tracking ID: 1001

Parcel Details

Sender: Aman Sharma

Receiver: Rahul Verma

Dispatch Address: Delhi Sorting Center

Delivery Address: MG Road Pune

Current Location: Delhi Sorting Center

Status: In Transit

Screenshots:

```
==============================
 Courier & Parcel Tracking System
==============================
1. Courier Staff
2. Customer
3. Exit
Select Role: 1

===== Courier Staff Menu =====
1. Register Parcel
2. Update Tracking
3. Confirm Delivery
4. View All Parcels
5. Back
Enter Choice: 1

Enter Sender Name: ayaan sharma
Enter Receiver Name: ptiti sharma
Enter Dispatch Address: MG road pune
Enter Delivery Address: Kandival mumbai

Parcel Registered Successfully!
Tracking ID: 1001

===== Courier Staff Menu =====
1. Register Parcel
2. Update Tracking
3. Confirm Delivery
4. View All Parcels
5. Back
Enter Choice: 2

Enter Tracking ID:  1001

Select Status Update
1. In Transit
2. Out For Delivery
Enter Choice: 1
Enter Current Location: navi mumbai

Tracking Updated!

===== Courier Staff Menu =====
1. Register Parcel
2. Update Tracking
3. Confirm Delivery
4. View All Parcels
5. Back
Enter Choice: 5

==============================
```

```
Tracking Updated!

===== Courier Staff Menu =====
1. Register Parcel
2. Update Tracking
3. Confirm Delivery
4. View All Parcels
5. Back
Enter Choice: 5

==============================
 Courier & Parcel Tracking System
==============================
1. Courier Staff
2. Customer
3. Exit
Select Role: 2

===== Customer Menu =====
1. Track Parcel
2. Back
Enter Choice: 1

Enter Tracking ID:  1001

Parcel Details
----------------------------
Sender: ayaan sharma
Receiver: ptiti sharma
Dispatch Address: MG road pune
Delivery Address: Kandival mumbai
Current Location: navi mumbai
Status: In Transit

===== Customer Menu =====
1. Track Parcel
2. Back
Enter Choice: 2

==============================
 Courier & Parcel Tracking System
==============================
1. Courier Staff
2. Customer
3. Exit
Select Role: 
```

```
===== Customer Menu =====
1. Track Parcel
2. Back
Enter Choice: 2

==============================
 Courier & Parcel Tracking System
==============================
1. Courier Staff
2. Customer
3. Exit
Select Role: 1

===== Courier Staff Menu =====
1. Register Parcel
2. Update Tracking
3. Confirm Delivery
4. View All Parcels
5. Back
Enter Choice: 2

Enter Tracking ID:  1001

Select Status Update
1. In Transit
2. Out For Delivery
Enter Choice: 2
Enter Current Location: goregoan

Tracking Updated!

===== Courier Staff Menu =====
1. Register Parcel
2. Update Tracking
3. Confirm Delivery
4. View All Parcels
5. Back
Enter Choice: 4

All Parcels
--------------------------------------------------------------------------------------------------
       ID         Sender       Receiver            Dispatch            Delivery       Location         Status
--------------------------------------------------------------------------------------------------
    1001   ayaan sharma   ptiti sharma        MG road pune     Kandival mumbai       goregoanOut For Delivery

===== Courier Staff Menu =====
1. Register Parcel
2. Update Tracking
3. Confirm Delivery
4. View All Parcels
5. Back
Enter Choice:
```

6. Problem Statement / Case Study Results and Conclusion.

Findings

The Courier & Parcel Tracking System successfully demonstrates how a simple program can be used to manage parcel information and track deliveries. The use of object-oriented programming and vectors allows the system to store and manage parcel records efficiently.

The menu-driven structure makes the system easy to use, and the tracking ID system allows customers to check parcel status quickly.

Conclusion

This case study shows how programming concepts learned in C++ can be applied to solve a real-world logistics problem. The system provides basic functionality for registering parcels, updating delivery status, and tracking parcels.

Although the system is simple and console-based, it provides a good foundation for building more advanced courier management systems.

Future improvements could include:

- Database storage for permanent records
- Graphical user interface (GUI)
- Online parcel tracking
- Automatic delivery notifications

Overall, the project demonstrates the practical use of programming techniques to create a useful application.

7. References

- C++ Programming Documentation – www.cplusplus.com
- GeeksforGeeks – C++ Programming Tutorials
- Course materials and lecture notes on Object-Oriented Programming
- Online resources related to courier tracking systems and logistics software